

Resume Classifier

Submission Report for Veridia Internship

Deadline: October 20, 2025

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Internship Program: Veridia Internship Program

Repository Link: <https://github.com/Gouravj410/ResumeClassifier.git>

Streamlit Link: <https://gouravj410-resumeclassifier-resume-app-ofnk61.streamlit.app/>

Project Overview :

Designed a Model that predicts the category of resume.

Tech Stack Used

Technology	Tools
Python Libraries	pandas, numpy, scikit-learn, matplotlib, seaborn, joblib
Web Framework	Streamlit
PDF Processing	PyPDF2
Version Control	Git & GitHub

Approach Followed :

- 1. Data Collection** - The dataset was taken from Kaggle's Resume Dataset, which contains resumes labeled into different job categories.
- 2. Data Preprocessing and Cleaning**
 - Removed unnecessary characters, punctuation, and HTML tags.
 - Converted text to lowercase.
 - Removed stopwords and performed lemmatization to clean the text.
 - Handled missing values and ensured all resumes were in text format.
- 3. Feature Extraction**
 - Used TF-IDF (Term Frequency–Inverse Document Frequency) vectorization to convert text into numerical features.
 - The TF-IDF vectorizer captures the importance of each word relative to the dataset.
- 4. Model Training**
 - Trained a machine learning classifier (such as Logistic Regression / SVM / Naive Bayes — depending on the code used) on the processed data.
 - The data was split into training and testing sets (e.g., 80-20 ratio).
- 5. Evaluation**
 - Calculated accuracy, precision, recall, and F1-score using `scikit-learn` metrics.
 - Achieved an overall accuracy of ****75.05 %**** on the test dataset.
 - Plotted the confusion matrix to visualize performance across categories.
- 6. Model Deployment**
 - The trained model and TF-IDF vectorizer were saved as `.pkl` files using Joblib.
 - A Streamlit web application was built where users can upload resumes in text or PDF format.
 - The app predicts the category/domain of the uploaded resume in real time.

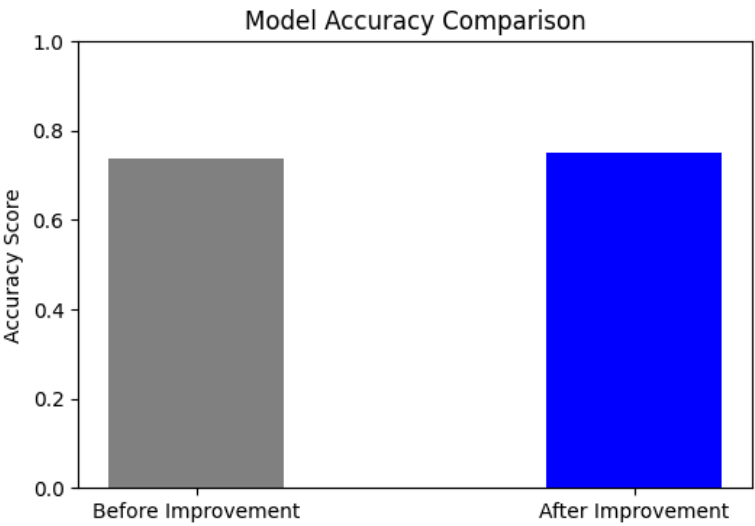
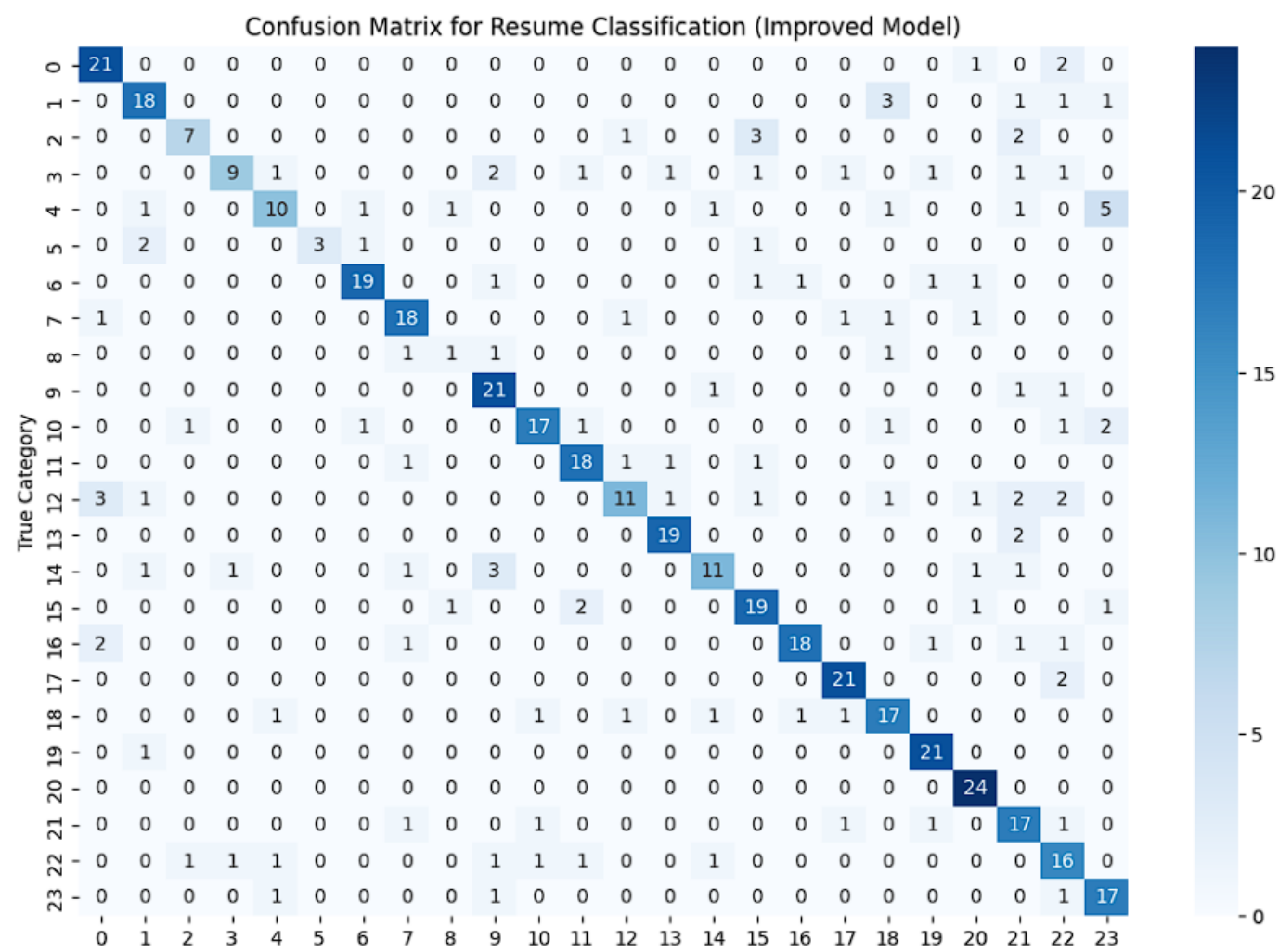
Results & Accuracy Metrics :

Model Accuracy: 75.05 %

- The confusion matrix and classification report showed that the model performs well for major categories like Data Science, HR, and Web Development.
- Minor misclassifications occurred between similar domains (e.g., Machine Learning vs. Data Science).
- The TF-IDF based text representation proved effective in handling varied resume structures.
- Overall, the model demonstrates reliable performance for automated resume classification.

Visualization Outputs:

1. Confusion Matrix — shows correct vs. incorrect predictions for each category.



2. **Classification Report** — includes precision, recall, and F1-score for all classes.

Improved Accuracy: 0.7505030181086519

	precision	recall	f1-score	support
ACCOUNTANT	0.78	0.88	0.82	24
ADVOCATE	0.75	0.75	0.75	24
AGRICULTURE	0.78	0.54	0.64	13
APPAREL	0.82	0.47	0.60	19
ARTS	0.71	0.48	0.57	21
AUTOMOBILE	1.00	0.43	0.60	7
AVIATION	0.86	0.79	0.83	24
BANKING	0.78	0.78	0.78	23
BPO	0.33	0.25	0.29	4
BUSINESS-DEVELOPMENT	0.70	0.88	0.78	24
CHEF	0.85	0.71	0.77	24
CONSTRUCTION	0.78	0.82	0.80	22
CONSULTANT	0.73	0.48	0.58	23
DESIGNER	0.86	0.90	0.88	21
DIGITAL-MEDIA	0.73	0.58	0.65	19
ENGINEERING	0.70	0.79	0.75	24
FINANCE	0.90	0.75	0.82	24
FITNESS	0.84	0.91	0.88	23
HEALTHCARE	0.68	0.74	0.71	23
HR	0.84	0.95	0.89	22
INFORMATION-TECHNOLOGY	0.80	1.00	0.89	24
PUBLIC-RELATIONS	0.59	0.77	0.67	22
SALES	0.55	0.70	0.62	23
TEACHER	0.65	0.85	0.74	20
accuracy			0.75	497
macro avg	0.75	0.72	0.72	497
weighted avg	0.76	0.75	0.74	497

3. **Streamlit Web App** — allows users to upload resumes (text/PDF) and get instant category predictions.



Resume Classifier App

Upload your resume (PDF) or paste text below to predict its category.

Upload Resume (PDF)



Drag and drop file here

Limit 200MB per file • PDF

Browse files

Or paste resume text here

Predict Category



Resume Classifier App

Upload your resume (PDF) or paste text below to predict its category.

Upload Resume (PDF)



Drag and drop file here

Limit 200MB per file • PDF

Browse files



GouravKishorJangid_Resume.pdf 136.7KB



PDF successfully loaded! You can now predict the category.

Or paste resume text here

Predict Category

Predicted Category: **DIGITAL-MEDIA**

Developer Information

Name: Gourav Jangid

Program: Veridia Internship Program

Date: October 20, 2025

"The future belongs to those who embrace technology wisely."